Kural2Action

Turning citizen voices into actionable civic intelligence

PROJECT PITCH | Gnani AI Great Indian AI Internship Challenge

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THE PROBLEM

Citizens may struggle to report local infrastructure issues when complaint processes require lengthy forms, structured descriptions, or location details they may not know how to provide. Reports can arrive incomplete and be difficult to organise for follow-up.

THE SOLUTION

Kural2Action is a voice-first civic complaint intake and monitoring prototype. A citizen describes an issue in Tamil or English; the application transcribes the audio, structures the available details, asks for missing information, and lets the citizen review and confirm the location before saving the report.

01 SPEAK	02 TRANSCRIBE	03 CLARIFY	04 CONFIRM	05 REVIEW	06 EXPLORE
Citizen records a complaint	Gnani Prisma STT	Extract and ask follow-up	Citizen confirms location	Duplicate & priority support	SQLite + map dashboard

WHAT THE PROTOTYPE DEMONSTRATES

Gnani AI voice: Prisma speech-to-text and Timbre text-to-speech.
Complaint structuring: keyword-based category, duration and preliminary severity extraction.
Citizen-in-the-loop: clarification and explicit location confirmation.
Decision support: similarity-based duplicate screening and transparent rule-based priority scoring; unassessed items are flagged for review.
Monitoring: Streamlit dashboard, charts, complaint records, map markers and heatmap when coordinates are available.

TECHNOLOGY

Python · Streamlit · Gnani Prisma API · Gnani Timbre API · SQLite · Pandas · Plotly · Folium · Geopy

RESPONSIBLE SCOPE

This is a working prototype, not a live municipal service. Reports are stored locally and are not submitted to government systems. Category and severity extraction, duplicate screening, and priority scores are preliminary decision-support signals; possible duplicates and unassessed complaints require human review. Location accuracy depends on available coordinates and citizen confirmation.

POTENTIAL IMPACT & NEXT STEPS

The prototype explores a more accessible way to capture regional-language civic reports and organise them for review. Next steps include evaluation with consented real-world data, stronger multilingual understanding, improved semantic matching and geocoding, privacy controls, and carefully governed integration with civic service workflows.

Listen to citizens. Understand the issue. Support civic action.

GitHub: https://github.com/Humaira3366/Kural2Action